AI Image Generator

1. Project Overview

Version 1.0
22 October 2026

Project owner: Frazile AI
Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Stable Diffusion XL + Real-ESRGAN, served locally with Gradio, with per-image GPU telemetry recorded in SQLite.

1. Project Overview

1.1 Purpose

AI Image Generator is a self-hosted application that turns a plain-English prompt into an image using the Stable Diffusion XL model, running entirely on a local NVIDIA GPU. No image data leaves the machine and there is no per-image cost.

The project was built in a single development session on 21 and 22 October 2026 and grew through five increments:

1. A command-line script that runs Stable Diffusion XL on the GPU.
2. A local web interface so a prompt can be typed and the result viewed in a browser.
3. A prompt language that replaces sliders and dropdowns: size, ratio, quality, style, count and seed are all read from the prompt text.
4. Per-image telemetry (timings, VRAM, temperature, power, clocks) shown in the UI and stored in a SQLite database with a History tab.
5. A Real-ESRGAN upscaling stage so that "4k" or "8k" in the prompt produces a genuine 3840 or 7680 pixel image.

1.2 Key capabilities

- Text-to-image generation with Stable Diffusion XL base 1.0 in 16-bit precision.
- Prompt-driven control of aspect ratio, exact resolution, orientation, quality, fidelity, art style, image count, seed, steps and guidance.
- Live interpretation line under the prompt box showing exactly what will be generated before the button is pressed.
- Tiled Real-ESRGAN upscaling to 2K, 4K, 8K or any exact size, with automatic centre crop.
- Full telemetry per image: inference time, steps per second, peak VRAM, GPU temperature and power before and after, clocks, fan, performance state, driver and library versions.
- Persistent history in SQLite with search, aggregate statistics, and click-to-view of any past image with its full record.
- Command-line tools for batch generation, upscaling an existing image, and querying the database.

1.3 Target environment

The application was developed and verified on the following machine. Other NVIDIA GPUs with at least 12 GB of VRAM will work; see the Requirements section for the full matrix.

Item	Value
GPU	NVIDIA GeForce RTX 3090, 24 GB VRAM, compute capability 8.6, 82 SMs
Driver	610.88
OS	Windows 11 Pro 10.0.26300
Python	3.12.0
CUDA runtime (via PyTorch)	12.6
Project folder	C:\Users\frazi\Documents\apps\ai-image-gen

1.4 Document map

Section	Contents
2. Requirements	Functional and non-functional requirements, hardware and software matrix
3. Architecture	Components, pipeline diagram, data flow, design decisions
4. Workflows	Setup, generation, upscaling, history and maintenance workflows step by step
5. Module reference	Every source file, its public functions and their contracts
6. Prompt language	Complete reference of words the parser understands
7. Database	Schema of generations.db, every column explained, example queries
8. Performance	Measured timings and resource use on the RTX 3090
9. Operations	Running, stopping, troubleshooting, known issues, backup
10. Change log and roadmap	What was built in which order, defects fixed, future work
Appendix	File tree, dependency list, command cheat sheet